BD-M

CORPORATE FINANCE -

10

Ch. ~~9~~ Cost of Capital

1. Which of the following is *not* considered a capital component for the purpose of calculating the weighted average cost of capital as it applies to capital budgeting?

   a. Long-term debt.
   b. Common stock.
   c. Short-term debt used to finance seasonal current assets.
   d. Preferred stock.
   e. All of the above are considered capital components for WACC and capital budgeting purposes.

2. For a typical firm with a given capital structure, which of the following is correct? All rates are after taxes.

   a. $r_d > r_s >$ WACC.
   b. $r_s > r_d >$ WACC.
   c. WACC $> r_s > r_d$.
   d. $r_s >$ WACC $> r_d$.
   e. None of the statements above is correct.

3. Which of the following statements is most correct?

   a. If a company's tax rate increases but the yield to maturity of its noncallable bonds remains the same, the company's marginal cost of debt capital used to calculate its weighted average cost of capital will fall.
   b. All else equal, an increase in a company's stock price will increase the marginal cost of common stock, r_s.
      $R = \frac{D_1}{\uparrow P_0}(\downarrow) + g \quad \therefore \downarrow$
   c. All else equal, an increase in interest rates will decrease the marginal cost of common stock, r_s.
   d. Answers a and b are correct.
   e. Answers b and c are correct.

4. Which of the following factors in the discounted cash flow (DCF) approach to estimating the cost of common equity is the *least* difficult to estimate?

   a. Expected growth rate, g.
   b. Dividend yield, D_1/P_0.
   c. Required return, r_s.
   d. Expected rate of return, $\hat{r}_s$.
   e. All of the above are equally difficult to estimate.

5. Which of the following statements is most correct?

a. Capital components are the types of capital used by firms to raise money. All capital comes from one of three components: long-term debt, preferred stock, and equity.
b. Preferred stock does not involve any adjustment for flotation cost since the dividend and price are fixed.
c. The cost of debt used in calculating the WACC is an average of the after-tax cost of new debt and of outstanding debt.
d. The opportunity cost principle implies that if the firm cannot invest retained earnings and earn at least r_s, it should pay these funds to its stockholders and let them invest directly in other assets that do provide this return.
e. The cost of common stock, r_s, is usually less than the cost of preferred stock.

6. Which of the following statements is most correct?

a. Suppose a firm is losing money and thus, is not paying taxes, and that this situation is expected to persist for a few years whether or not the firm uses debt financing. Then the firm's after-tax cost of debt will equal its before-tax cost of debt.
b. The component cost of preferred stock is expressed as $r_{ps}(1 - T)$, because preferred stock dividends are treated as fixed charges, similar to the treatment of debt interest.
c. The reason that a cost is assigned to retained earnings is because these funds are already earning a return in the business; the reason does not involve the opportunity cost principle.
d. The bond-yield-plus-risk-premium approach to estimating a firm's cost of common equity involves adding a subjectively determined risk-premium to the market risk-free bond rate.
e. All of the statements above are false.

7. An analyst has collected the following information regarding Christopher Co.:

- The company's capital structure is 70 percent equity, 30 percent debt.
- The yield to maturity on the company's bonds is 9 percent.
- The company's year-end dividend is forecasted to be $0.80 a share.
- The company expects that its dividend will grow at a constant rate of 9 percent a year.
- The company's stock price is $25.
- The company's tax rate is 40 percent.
- The company anticipates that it will need to raise new common stock this year. Its investment bankers anticipate that the total flotation cost will equal 10 percent of the amount issued. Assume the company accounts for flotation costs by adjusting the cost of capital. Given this information, calculate the company's WACC.

a. 10.41%
b. 12.56%
c. 10.78%
d. 13.55%
e. 9.29%

8. A company's balance sheets show a total of $30 million long-term debt with a coupon rate of 9 percent. The yield to maturity on this debt is 11.11 percent, and the debt has a total current market value of $25 million. The balance sheets also show that that the company has 10 million shares of stock; the total of common stock and retained earnings is $30 million. The current stock price is $7.5 per share. The current return required by stockholders, r_S, is 12 percent. The company has a target capital structure of 40 percent debt and 60 percent equity. The tax rate is 40%. What weighted average cost of capital should you use to evaluate potential projects?

a. 8.55%
b. 9.33%
c. 9.36%
d. 9.87%
e. 10.67%

9) On January 1, the total market value of the Hoppenmeimer Company was $20 million. During the year, the company plans to raise and invest $10 million in new projects. The firm's present market value capital structure, shown below, is considered to be optimal. Assume there is no short-term debt.

Debt	$10,000,000
Common Equity	10,000,000
Total Capital	$20,000,000

New 30-year bonds will have a 7% stated rate, are discounted $59.36 for each $1,000 par value bond when issued, and pay interest semi-annually. Stockholder's required rate of return is consists of a 5% dividend yield to the investor based on the next dividend paid out of $1.25 a share, and an expected constant growth rate of 8%. Flotation costs to the issuer are $2 a share. The marginal corporate tax rate is 40%.

a. To maintain the present capital structure, how much of the new investment (in millions) must be financed by common equity. $5,000,000
b. What is the net price per share to the company if they use external equity and issue stock to finance the business? How many new shares must be issued? $23.00 price; NEW SHARES = 217,392
c. Now assume that there is sufficient cash flow so that Hoppenmeimer can maintain its target capital structure without issuing additional shares of equity. What is the WACC? 8.75%

10) Schnidt Company has this book value balance sheet:

Current Assets	$45,000,000	Notes Payable	$ 15,000,000
Fixed Assets	75,000,000	L/T Debt	45,000,000
		Common Equity:	
		Common Stock (1.5 million shares)	1,500,000
		RE	58,500,000
Total Assets	$120,000,000	Total Liab & OE	$120,000,000

The interest rate on the notes payable is 10%, the same as new bank loans. The long-term debt consists of 45,000 bonds with 20 years to maturity. Each bond has a par value of $1,000 and an annual coupon rate of 6%. The going rate of interest on new long-term debt of similar risk and maturity is 10%. The common stock finished at $30 per share at the end of today's trading.

a. What is the market value of the firm's long-term debt? $29,675,585.
b. What is the market value weight of common equity in the firm's capital structure if short-term debt is included? 50.18%

COST CAPITAL B

PROB SET

7)

$$WACC = (wtd)(kd)(1-t) + (wtcom)(K_{CE})$$

$$= (.3)(.09)(1-.4) + (.7)(.1256)$$

$$.0162 + .08792$$

$$K_e = \frac{D_1}{P_0(1-f)} + g$$

$$= 10.41\%$$

$$\frac{.8}{25(1-.1)} + .09$$

$$\frac{.8}{22.5} + .09$$

$$= .1256$$

COST CAPITAL
PROB SET

8) $WACC = (w_d)(k_d)(1-t) + (w_e)(k_e)$ COST ⇨ Return

$(.4)(.1111)(1-.4) + (.6)(.12)$

Target

$.0267 + .072 = .0987$

or

9.87%

Ch. 11 Cost of Capital

9)

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CH 10 COST OF CAPITAL
PROB SET

#9)

MV = \$20,000,000

Debt	\$10,000,000	Tax = 40%
Comn Eqty	10,000,000	
Total capital	\$20,000,000	

BONDS

30 yrs
Stated rate = 7% semi-annual
Par Value = <\$1000>
Discount = \$59.36

STOCK

Div Yld = 5%
D_1 = \$1.25
g = 8%
Flot. costs = \$2/share

a) $\frac{\text{Comn Eqty}}{\text{Total Cap}} = \frac{\$10 \text{ mill}}{\$20 \text{ mill}}$ = 50% Eqty Financed

x \$10 mill need to be raised

= \$5,000,000 Comn Eqty Needed

b) $P_0 = \frac{D_1}{k-g} = \frac{\$1.25}{k-.08}$

Need Req'd Return K

but know Div Yld + Const. Growth = Req'd Return

5% + 8% = 13%

so: $P_0 = \frac{\$1.25}{.13-.08}$ & P_0 = \$25.

BUT ⇒ want NET price to company if NEW EQTY
so adjust for flotation

P_0 = \$25 − \$2 Flot. costs

P_0 = \$23.

$\frac{\$5,000,000 \text{ NEW EQTY}}{\$23}$

217,392 shs issued

PROB SET

#9c)

DEBT

Before Tax Cost of Debt

$N = 30$ yrs $\times 2 = 60$ semi-annual period

Stated (Coupon Rate) $= 7\%/2 = 3.5\% \times \$1000 = \$\langle 35 \rangle = PMT$

Par Value $= FV = \$\langle 1{,}000 \rangle$

$-$Discount $= -\ 59.36$

$=$ Issue (Selling) Price $\$\ 940.64 = PV$

WANT ; $YTM = i = 3.75\% \times 2 = 7.50\%$

B/4 Tax $(1 - tax)$ = After tax

$7.50\,(1 - .4) = \underline{\underline{4.5\%}}$ A/T cost

$$WACC = \left(\begin{matrix}Wt\\ Debt\end{matrix}\right)\left(\begin{matrix}A/tax\\ cost\ Debt\end{matrix}\right) + \left(\begin{matrix}Wt\\ Eqty\end{matrix}\right)\left(\begin{matrix}COST\\ EQTY\end{matrix}\right)$$

$$\frac{DEBT}{T.Cap}\left(\frac{10\ MILL}{20\ MILL}\right)(4.5\%) + \left(\frac{\overset{EQTY}{10\ MILL}}{\underset{T.Cap}{20\ MILL}}\right)(13\%)$$

$$WACC = 2.25\% + 6.5\% = 8.75\%$$

#10

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a. What is the market value of the firm's long-term debt?
b. What is the market value weight of common equity in the firm's capital structure if short-term debt is included?

L/T DEBT

FV = 1000
PMT = 6% x 1000 = 60
N = 20
i = 10%
PV = ? = $659.46
x 45,000 bonds
$29,675,585
= MV of bonds

MV of COMMON EQTY

shs 1,500,000
Mrt. Pr. $30

			M.V. WEIGHT
C. Stk MV =	45,000,000 ÷	89,675,585 =	50.18 %
Debt MV =	29,675,585 ÷	" =	33.09%
N/Pble =	15,000,000 ÷	" =	16.73%
TOTAL CAP. =	89,675,585		100 %

#10

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x 45,000 bonds

$29,675,585

MV of bonds

MV of COMMON EQTY

shs 1,500,000

Mrt. Pr. $30

		M.V. WEIGHT
C. Stk MV = 45,000,000	÷ 89,675,585 =	50.18%
Debt MV = 29,675,585	÷ " =	33.09%
N/Pble = 15,000,000	÷ " =	16.73%
TOTAL CAP. = 89,675,585		100%

CH 9
COS OF CAPITAL

C
2) D
3) A
4) B
5) D
6) A
7) A
8) D
9) a) $5 MILL
b) $23.00 price ; 217,392 NEW SHARES
c) 8.75%
10) a) $29,675,585
b) 50.18%

CH